

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Annexure A Scenario-Based

Code of Conduct:

I Ashwin V-(192424345) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V Ashwin

Annexure A Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based

on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

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1. Smart Campus Network (192.168.100.0/24)

Given block: 192.168.100.0/24

Total addresses: 256 (192.168.100.0 – 192.168.100.255)

School of Computing → /25

Subnet mask: 255.255.255.128

- Network Address: 192.168.100.0
- Broadcast Address: 192.168.100.127
- Valid Host Range: 192.168.100.1 – 192.168.100.126
- Usable Hosts: 126

School of Electronics → /26

Subnet mask: 255.255.255.192

- Network Address: 192.168.100.128
- Broadcast Address: 192.168.100.191

- Valid Host Range: 192.168.100.129 – 192.168.100.190
- Usable Hosts: 62

School of Mechanical Engineering → /27

Subnet mask: 255.255.255.224

- Network Address: 192.168.100.192
- Broadcast Address: 192.168.100.223
- Valid Host Range: 192.168.100.193 – 192.168.100.222
- Usable Hosts: 30

Remaining Unused Range Remaining
addresses:

192.168.100.224 – 192.168.100.255

Total unused addresses: 32

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid range of host IP addresses.

3. Calculate the number of usable hosts in each subnet.

4. Identify the unused IP address range remaining within the allocated IP block.

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2. Software Company Network (10.10.0.0/16) Network range:

10.10.0.0 – 10.10.255.255

Software Development → /24

- **Network Address: 10.10.0.0**
- **Broadcast Address: 10.10.0.255**
- **Host Range: 10.10.0.1 – 10.10.0.254**
- **Usable Hosts: 254**

Quality Assurance → /25

- **Network Address: 10.10.1.0**
- **Broadcast Address: 10.10.1.127**
- **Host Range: 10.10.1.1 – 10.10.1.126**
- **Usable Hosts: 126**

Human Resources → /26

- **Network Address: 10.10.1.128**
- **Broadcast Address: 10.10.1.191**
- **Host Range: 10.10.1.129 – 10.10.1.190**
- **Usable Hosts: 62**

Executive Management → /27

- Network Address: 10.10.1.192
- Broadcast Address: 10.10.1.223
- Host Range: 10.10.1.193 – 10.10.1.222
- Usable Hosts: 30

Remaining Unused Range Unused addresses:

10.10.1.224 – 10.10.255.255

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

- 1. Convert the given IPv6 address into its compressed notation.**
- 2. Expand the compressed address back into its full hexadecimal representation.**
- 3. Identify the network prefix using the /64 prefix length.**
- 4. Calculate the total number of host addresses supported within the subnet.**
- 5. the total number of possible host addresses available within this network.**

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3. IPv6 Hospital Network Given:

2001:0db8:abcd:0000:0000:0000:0000/64

1. Compressed Notation

Remove leading zeros and compress consecutive zero groups:

2001:db8:abcd::/64

2. Expanded Form

Compressed address:

2001:db8:abcd::

Expanded:

2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network Prefix

Prefix length: /64

Network portion:

2001:0db8:abcd:0000::/64

First 64 bits identify the network.

4. Total Host Addresses Supported IPv6 has 128 bits. Host bits:

128 – 64 = 64 bits

Number of addresses:

$2^{64} = 18,446,744,073,709,551,616$ addresses

5. Possible Host Addresses Available

Available host addresses:

18,446,744,073,709,551,616

(IPv6 normally does not subtract network and broadcast addresses like IPv4.)

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address: 192.168.2.45

As the routing engineer, determine:

- 1. The destination subnet to which the packet belongs.**
- 2. The matching routing table entry.**
- 3. The next-hop network/interface used to forward the packet.**
- 4. The default route that should be used if the destination does not match any routing table entry.**

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4. ISP Routing Table Routing

entries:

- 192.168.1.0/24**
- 192.168.2.0/24**
- 192.168.3.0/24 Destination IP:**

192.168.2.45

1. Destination Subnet

The IP belongs to:

192.168.2.0/24

2. Matching Routing Entry

Matching entry:

192.168.2.0/24

3. Next-Hop Network/Interface

The router forwards the packet through the interface connected to:

192.168.2.0/24 network

(The exact next-hop address depends on the router configuration.)

4. Default Route

If no route matches:

0.0.0.0/0

(Default gateway route)

5. A company establishes two office departments connected through a Layer3 router. The allocated subnet information is as follows:

Administration LAN → 192.168.10.0/26

Finance LAN → 192.168.10.64/26

The router interfaces are configured with:

Administration Gateway → 192.168.10.1

Finance Gateway → 192.168.10.65

As the network administrator, perform the following tasks:

- 1. Determine the network address and broadcast address for each LAN.**
- 2. Identify the valid host IP address range.**
- 3. Suggest suitable IP addresses for three computers in each LAN.**
- 4. Identify the default gateway that hosts in each LAN should use for communication.**

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192.168.10
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Example computers: • PC1: 192.168.10..... 2

• PC2: 192.168.10..... 3

• PC3: 192.168.10..... 4

5. Two Office LANs

Administration LAN → 192.168.10.0/26

Subnet mask:

255.255.255.192

- Network Address: 192.168.10.0
- Broadcast Address: 192.168.10.63 • Host Range: 192.168.10.1 – 192.168.10.62
- Usable Hosts: 62 Gateway:

Default Gateway:

192.168.10.1

Finance LAN → 192.168.10.64/26

Subnet mask:

255.255.255.192

- Network Address: 192.168.10.64
- Broadcast Address: 192.168.10.127 • Host Range: 192.168.10.65 – 192.168.10.126
- Usable Hosts: 62 Gateway:

192.168.10.65

Example computers:

- **PC1: 192.168.10.66**
- **PC2: 192.168.10.67**
- **PC3: 192.168.10.68**

Default Gateway:

192.168.10.65